

PROBLEM SCOPING

❖ Placement prediction system

University Placement Cell

- Notices that many students are unable to secure placements before graduation.
- Placement rate has decreased in recent years.
- Management wants to improve placement outcomes and student employability.

They approach an AI team/organization

- Asked to build an AI model/system to identify students who are likely to get placed.
- Also identify students who may need additional training and support.

Solution :

Understand the Business Problem



Problem statement



Need of AI



IDENTIFY STAKEHOLDERS



Defining the task of AI



DEFINING SUCCESS METRICS :



DEFINING CONSTRAINTS



ASSUMPTIONS AND RISKS

Step 1 : Understand the Business Problem

Initial statement:

"Many students are not getting placed."

Need to uncover the actual problem and define measurable objectives.

Questions for Stakeholders

To Placement Department

- What is the current placement rate?
- When has the placement rate decreased?
- Which departments have the lowest placement rates?
- What skills are most demanded by recruiters?

To Academic Department

- What academic factors affect placements?
- Is CGPA strongly related to placement success?
- How do attendance and academic performance impact placements?
- Which students are most at risk of remaining unplaced?

To Students

- Have students completed internships?
- How many technical projects have they completed?
- What programming skills do they possess?
- Are they interested in higher studies or jobs?
- What challenges do they face during placements?

To Management

- Why is improving placement rate important?

- What is the target placement percentage?
- What budget is available for training programs?
- How will the success of the AI system be measured?

Suppose the University Provides Following Reports

- Current placement rate: **72%**
- Desired placement rate: **90%**
- Most unplaced students have CGPA below **6.5**
- Students with internships have higher placement chances.
- Communication and aptitude skills strongly affect placement outcomes.
- Training cell can support around **300 students per semester**.
- Early identification of at-risk students can improve placement rates significantly.
-

Step 2 :Problem statement

Identify final-year students who are at high risk of remaining unplaced at least one semester before campus placements so that training and placement officers can provide targeted support.

- **What?**
Predict placement chances of students.
- **When?**
One semester before placement season.
- **Why?**
Enables early intervention and training.
- **Outcome:**
Increase placement rate and improve student employability.

Step 3 : Need of AI :

Can the problem be solved without AI?

If CGPA > 8.5 and Internship >= 1, then student is likely to be placed.

Why AI is needed ?

Placement depends on many factors:

1. CGPA
 2. 10th Percentage
 3. 12th Percentage
 4. Internships
 5. Projects
 6. Technical Skills
 7. Aptitude Score
 8. Communication Skills
 9. Resume Quality
 10. Mock Interview Performance
- **Patterns are complex.**
 - **Multiple factors interact.**
 - **Manual rules are insufficient.**

- **WITHOUT AI**

Students → Placement Season → Many Unplaced Students

- **WITH AI**

Students
↓
Placement Prediction System
↓
High-Risk Students Identified
↓
Targeted Training
↓
Improved Placement Rate

STEP 4: IDENTIFY STAKEHOLDERS

- *University Management:*
Improve placement percentage.

- *Students:*
Receive timely guidance and training.
- *Placement Cell:*
Identify students needing support.
- *Training Team:*
Provide focused training.
- *Faculty Members:*
Improve student outcomes.
- *Recruiters:*
Find suitable candidates.
- *IT Department:*
Deploy and maintain the system.

Step : 5 Defining the task of AI

Machine Learning Formulation

. Classification Problem

Output:

Placed , Not Placed , Binary Classification Problem

. Risk Scoring

Probability of Placement:

0% to 100%

Example:

- Student A = 92%
- Student B = 35%
- Student C = 78%

. Ranking Problem

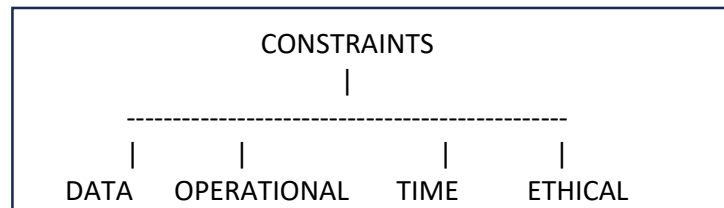
Which students should receive training first?

- ✓ Rank 1 = 20% placement chance
- ✓ Rank 2 = 25% placement chance
- ✓ Rank 3 = 30% placement chance

Step 6: DEFINING SUCCESS METRICS :

- Business Metrics
- Increase placement rate from 72% to 90%.
- Improve student employability.
- Better utilization of training resources.
- Technical Metrics
- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Target: Accuracy > 85% , Recall > 80%

Step 7 : DEFINING CONSTRAINTS



- **Operational Constraints**
 - Training team can support only 300 students per semester.
 - Limited training budget.
-
- **Time Constraints**
 - Predictions must be available before placement season.
-
- **Data Constraints**
 - Internship details may be missing.
 - Aptitude scores may be missing.
 - Communication skill ratings may be unavailable.
 - Resume quality data may not exist.
-
- **Ethical Constraints**
 - Students should not be unfairly judged.
 - Predictions should support decisions, not replace them.
-
- **Student privacy must be protected.**
 - Can predictions negatively affect students?
 - Can students appeal decisions?

Step 8 : ASSUMPTIONS AND RISKS

- **Assumptions**
 - Historical placement data reflects future trends.
 - Student records are accurate.
 - Training programs improve placement chances.
- **Risks**
 - Missing data.
 - Biased historical records.
 - Fewer companies visiting campus.
 - New technologies changing skill requirements.

Statistical Approach

